INTEGRATION OF COMMUNICATION AND SAR RADAR BASED ON OFDM WITH CHANNEL ESTIMATION IN HIGH SPEED SCENARIO

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Abstract— **With the development of science and requirement, Integration of Communication and Radar with properties of more functions, lower buck and safety is becoming a trend and draws more attention of researchers. Current researches are focusing on communication or radar respectively or just focusing on signal generation. This paper proposes a solution of integrated system for SAR radar communication based on OFDM techniques in high speed scenario. The receiver can reconstruct the reference signal which is adopted for radar pulse compression procedure by communication system directly. In high speed scenario, channel response changes fast, comb type pilots and channel estimation are added to equalize the channel response to reduce bit-error-rate in communication system. Finally, the simulation are used to validate the performance of the proposed algorithm.**

Index Terms—**Integration of Communication and SAR Radar, OFDM, Channel Estimation, SAR, Communication.**

I. INTRODUCTION

THE application of radar systems and communication systems is very common in battlefield environments. Usually, both systems are used separately and independently. However, the independent use of radar and communication equipment result in reduced mobility, wasted spectrum resources, and impaired individual performance [1]. We know that radar systems and communication systems are largely similar in structure, such as transmitters, receivers, and processing hardware. In [2] [3], radar and communication are capable of sharing spectrum. Therefore, the development of radar and communication integration systems has become more and more concerned. In [4], it is proposed the orthogonal frequency-division multiplexing (OFDM) SAR imaging, where a sufficient cyclic prefix (CP) is added to each OFDM pulse. A kind of orthogonal frequency diversion multiplexing (OFDM) chirp diverse waveform is presented for SAR imaging in [5]. In [6], an integrated system combining multiple input multiple output (MIMO) radar and orthogonal frequency division multiplexing (OFDM) communication is designed. There is no sufficient research on the whole integrated system especially in high speed scenario. This paper proposes a solution in bistatic platforms with high speed scenario.

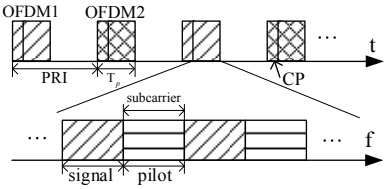
II. SYSTEM MODEL

In order to build the system model of integration system, some key technologies should be discussed first. We all know that electromagnetic wave travels from transmitter to receiver in both Radar and Communication, and the differences between them are signal generation and signal processing. In communication system, the receiver does not know the symbols from the transmitter, so the receiver demodulates and decodes the received signal to recover symbols. Whereas in Bistatic Radar system, the receiver needs the reference signal to make further processing such as pulse-compression. In signal generation, communication system modulates symbols on signals which makes it random. On the contrary, Radar system transmits a certain signal. So we transmit a communication signal on the radar system, and on the receiver, the signal is demodulated by communication system which will be used as the reference signal for radar signal processing.

In this section, we first analyze the signal model and then propose an algorithm for signal processing of integration system.

A. Signal Model

In this paper, the transmitter and receiver can move respectively, whose instantaneous coordinates are (x_t, y_t, z_t) and (x_r, y_r, z_r). In bistatic side-looking geometry, the transmitter and receiver coordinates are not collocated. Considering an OFDM signal with N subcarriers, whose band is B Hz. In the matrix form, the OFDM signal can be expressed as $\boldsymbol{S} = [S_0, S_1, \cdots, S_{N-1}]$. In $\boldsymbol{S}$ some frequencies are treated as pilots and the others are treated as communication symbols. Supposing the symbols of communication are $\boldsymbol{S}_D$, while the symbols of pilot are $\boldsymbol{S}_P$. Then we can make a combination of two signals to form signal $\boldsymbol{S}$ of integration.



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Fig. 1 signal model

For symbols of communication, the symbols are coded via 2DPSK. After symbols are coded, the $\exp(\cdot)$ function is used to generate complex envelope of frequency for OFDM signal. For pilots, we choose a signal that has great correlation ability. So Zadoff-Chu sequence is needed.

After inverse fast Fourier transform (IFFT) of $\boldsymbol{S}$. The OFDM signal of time-domain is

$$s(n)=\frac{1}{\sqrt{N}}\sum_{k=0}^{N-1}S_k\exp\left(j2\pi nk/N\right) \tag{1}$$

where n is discrete time after sampling processing. The sampling rate is $f_s=B$ in the condition of complex signal and OFDM signal. Then cyclic prefix should be added into the time-domain signal. The CP length is set according to the scenario. The length of CP is discussed in [4]. In another prospect, when we get echo of the transmitted signal, after CP is removed, the effective signal should contains all the information of the given scenario. After CP is added, the time-domain signal can also rewrite as follows

$$s(n)=\frac{1}{\sqrt{N}}\sum_{k=0}^{N-1}S_k\exp\left(j2\pi nk/N\right)\quad n\in[0,N+N_{GI}] \tag{2}$$

Then this signal can be modulated to the RF and transmitted. After demodulation to base band, the echo of m th range cell can be rewritten as

$$\begin{aligned}u_m(n,\eta)=&\,g_m\varepsilon_\alpha(\eta)\exp\left\{\frac{-j4\pi f_cR_m(\eta)}{c}\right\}\\&\times\frac{1}{\sqrt{N}}\sum_{k=0}^{N-1}S_k\exp\left(\frac{j2\pi k}{N}\left(n-\frac{2R_m(\eta)}{c}\right)\right)\\&+\omega(n,\eta)\quad n\in\left[\frac{2R_m(\eta)}{c},\frac{2R_m(\eta)}{c}+N+N_{GI}\right]\end{aligned} \tag{3}$$

where, g_m contains all the information of m th range cell, $\varepsilon_\alpha(\eta)$ is azimuth envelop, $R_m(\eta)$ is instantaneous range between the integration system and target.

For all range cells, adding all the echoes together, all the echoes can be deduced as follow:

$$u(n,\eta)=\sum_m u_m(n,\eta) \tag{4}$$

The m th range cell is $R_m(\eta)=R_0(\eta)+\Delta R_m$ Supposing all the echoes are aligned to the reference range cell. After alignment we just need to consider ΔR_m.

For one η, received echo in discrete form is $\boldsymbol{u}=[u_0,\cdots,u_{M-1},u_M,\cdots,u_{N-1}]$. Then the CP is needed to be removed. The remaining N sampling result is $\boldsymbol{u}=[u_M,\cdots,u_{N-1}]$

B. Signal Processing

In this paper, we combine radar system and communication system together. Our aims are listed as follow:

One, we need to reduce the buck of two systems, and make the system more functional.

Two, we can reconstruct reference signal via symbols decoded by communications which means the receiver does not need a copy-of-original-signal. So this method can be applied into bistatic scenario especially. As mentioned before, the procedures of signal processing can be shown as Fig. 2.

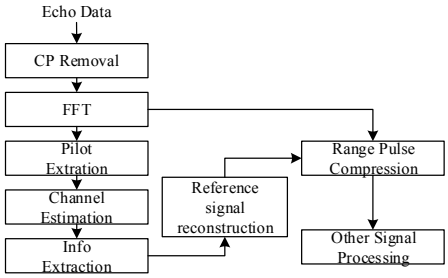


Fig. 2 signal processing

From Fig. 2, we make procedure of communication first, and then make procedure of radar. In order to get more accurate symbols of communication, channel estimation is needed to make an equalization of channel.

1) Communication Procedure

After CP is removed, we get the echo samples $\boldsymbol{u}$ in discrete form. Then fast Fourier transform (FFT) is carried on to achieve the frequency-domain result.

$$\boldsymbol{U}=\mathrm{FFT}\{\boldsymbol{u}\} \tag{5}$$

We make U_k represent the k th element of $\boldsymbol{U}$. Because we know the locations of pilots and the values of pilots. If we treat the propagation from transmitter to receiver as a linear filter, and its pulse response is $h(t)$ in consecutive form. So the pilots $\boldsymbol{S}_p$ and received pilots $\hat{\boldsymbol{S}}_p$ have the following equation:

$$\hat{\boldsymbol{S}}_p=\boldsymbol{H}_p(f)\times\boldsymbol{S}_p \tag{6}$$

In this equation, $\boldsymbol{H}_p(f)$ is frequency response at the locations of pilots. So we can get a specific response of pilots as

$$\boldsymbol{H}_p(f)=\hat{\boldsymbol{S}}_p\times\boldsymbol{S}_p^{\dagger} \tag{7}$$

Then interpolation is carried on to make the whole estimation of channel. In order to carry on interpolation, we first transform $\boldsymbol{H}_p(f)$ to the time-domain $\boldsymbol{h}_p(t)$. Then zeroes are padded to the front and the rear of $\boldsymbol{h}_p(t)$ can be deduced as:

$$\hat{\boldsymbol{h}}(t)=\left[\overbrace{\underbrace{0,\cdots,0}_{\frac{N-N_p}{2}},h_1,\cdots,h_p,\underbrace{0,\cdots,0}_{\frac{N-N_p}{2}}}^{N}\right] \tag{8}$$

Then we carry on fast Fourier transform to turn time-domain result into frequency-domain result $\hat{\boldsymbol{H}}(f)$. Then we can equalize the echo signal by $\hat{\boldsymbol{H}}(f)$, that means :

$$\hat{\boldsymbol{U}}=\boldsymbol{U}/\hat{\boldsymbol{H}}(f) \tag{9}$$

After equalization, we can decode the communication signal.

Once symbols in communication are decoded. The reference signal can be reconstructed. The way to reconstruct reference signal $\boldsymbol{S}_r$ is the same as described in signal model.

2) Radar Procedure

As mentioned before, we carried on the fast Fourier transform of $\boldsymbol{u}$, and reference signal $\boldsymbol{S}_r$ reconstructed Pulse compression in range-domain can be executed next.

According to typical pulse compression. In frequency domain, let the echo data of frequency times conjecture of the

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reference signal in frequency, then inverse fast Fourier transform is carried on to accomplish the pulse compression. According to the algorithm of SAR imaging, in this paper, we do not adopt IFFT in this step, just make a multiplication, and we get:

$$\hat{\boldsymbol{D}}_k = \boldsymbol{U} \times (\boldsymbol{S}_r)^* \tag{10}$$

Where $(\cdot)^*$ represents conjecture. After the pulse compression, SAR processing can be achieved.

So after combination of communication and radar, we can get more functions in integrated system. Meanwhile the receiver does not need a copy-of- signal to accomplish radar procedure.

III. SYSTEM ANALYSIS

When pilots are inserted into data symbols, the locations of pilots can not carry other information anymore. This will degrade data rate of communication. In order to weigh the loss, we define $L_p = N_p / N$, where N_p is the number of pilots. Data rate for communications are degraded, but this loss has no impact on radar processing.

Both receiver and transmitter know locations and values of pilots. The receiver can reconstruct the whole transmitted signal approximately. But some loss will also appears because the receiver can not recovery all the data symbols that transmitted in communication. This will result in the reference signal can not equal to the transmitted signal. The value to weigh the loss is bit-error-rate. If ber is high enough, radar processing can not be achieved. So we must control ber at a lower level.

When transmitter and receiver are move fast, in order to decrease ber, the Doppler between integration system and targets become higher, meanwhile. This will result in the coherence time become very small. In other words, duration of data symbols can greater than coherence time, the channel will turn into fast fading, which has great impacts on integration system. Besides, if coherence band become small, this will pull wide-band signal to frequency-selective-fading. We must design band of signal correspond to the scenario. In order to reduce the impact, we need to carry on channel estimation in every pulse.

In high speed circumstance, channel estimation is needed in radar and communication procedure. Pilots that are sent by transmitter are $\boldsymbol{S}_p$, while pilots that receiver gets is $\hat{\boldsymbol{S}}_p$. When signal propagates from transmitter to receiver. Additive-white-noise is added to the signal. At the step of channel estimation, in time domain we can get:

$$\hat{\boldsymbol{s}}_p(t) = \boldsymbol{h}_p(t) * \boldsymbol{s}_p(t) + \boldsymbol{\omega}_p \tag{11}$$

In frequency domain we also get:

$$\begin{aligned} \hat{\boldsymbol{S}}_p(f) &= \boldsymbol{H}_p(f)\boldsymbol{S}_p(f) + \boldsymbol{W}_p \\ \boldsymbol{H}_p(f) &= \hat{\boldsymbol{S}}_p \times \boldsymbol{S}_p^{\dagger} - \boldsymbol{W}_p \times \boldsymbol{S}_p^{\dagger} \end{aligned} \tag{12}$$

We can take 2-norm of equation above, this will turn into:

$$\begin{aligned} &\left\|\boldsymbol{H}_p(f)\right\| = \left\|\hat{\boldsymbol{S}}_p \times \boldsymbol{S}_p^{\dagger}\right\| - \left\|\boldsymbol{W}_p \times \boldsymbol{S}_p^{\dagger}\right\| \\ &\Rightarrow \left\|\boldsymbol{H}_p(f)\right\| = \left\|\hat{\boldsymbol{S}}_p\right\| \times \left\|\boldsymbol{S}_p^{\dagger}\right\| - \left\|\boldsymbol{W}_p\right\| \times \left\|\boldsymbol{S}_p^{\dagger}\right\| \end{aligned} \tag{13}$$

In this equation, $\left\|\boldsymbol{S}_p^{\dagger}\right\| > 1$, the additive-white-noise will become larger than the original one. For $\left\|\boldsymbol{S}_p^{\dagger}\right\| = 1/\left\|\boldsymbol{S}_p\right\|$, and well will let $\left\|\boldsymbol{S}_p\right\| \geq 1$. If we adopt Zadoff-Chu and other constant-module sequence, this phenomenon will be eased. After interpolation the whole noise will degrade.

The same as LFM and other signals that have good ability in resolution. This signal contains communication symbols. But the band of signal is also spread in a certain frequency. This make high resolution possible. If band of the integration signal is B, then the resolution is $\rho_r = c/2B$, where c is the speed of light. If we make the band as wide as possible under the constraints. We can get high resolution also. From the analysis before, ber causes the reference signal is not the same as the original one. This will cause pulse compression loss in radar processing. The side effect is value of side-lobe increases but this will not impact resolution for different targets in low ber condition.

We all know that at the procedure of imaging, phase between different pulse in frequency-domain can not change very high. In practice this value follows the constraint of $\Delta\phi < \pi/4$ in order to guarantee the quality of image. In order to achieve this goal, we need to design the signal that is transmitted. If signal has a good orthogonal ability, the impact of phase between adjacent pulses will be decreased. As Zadoff-Chu sequence has good orthogonal ability between different Zadoff-Chu sequences. But for any other sequences there is no guarantee for the orthogonality. Besides we choose 2DPSK as a specific way to code data symbols. After coded and pilots insertion, result of auto-correlation only has main lobe, this is what we need in radar processing.

IV. SIMULATIONS

The aim of this section is to verify theorem that proposed in the front of this paper. Geometry in simulation is shown in Fig1. There are 4×4 points uniformly located in 500m×500m area. In range domain, there are 4 points, and 4 points in azimuth.

According to the constraints, some parameters are listed as following: PRT=100us, the pulse duration time tau=30us, the band of signal is B=20MHz, the carrier-frequency $f_c = 10\text{GHz}$, velocity of transmitter is $v_t = 300$ m/s and its direction is 15.5rad, velocity of receiver is $V_r = 300$ m/s and its direction is 22.5rad, the original positions for transmitter and receiver are $(0\text{m}, 10000\text{m}, 5000\text{m})$ and $(0\text{m}, 15000\text{m}, 5000\text{m})$, the sampling frequency $f_s = 30$ MHz, the number of subcarrier is 900, $\Delta f = 33.3$ KHz, which spread all sampling frequency, and only 600 subcarrier is used for efficient transmission to meet Band. Let there are 1000 pulses in azimuth. So the Doppler is less than frequency spacing. In extreme condition, the coherence time is $0.423 * 3e8 / 10e10 / 600 \approx 21us$. So the pulse duration is also satisfying to the constraints. Corresponding to the scenario, the band satisfies the constraints also. And the spacing of adjacent pilots are 4. We set SNR at receiver is form 0dB to 20dB. Then we can carry on simulation.

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A. Channel Estimation Results

In this circumstance, estimation error is shown in Fig. 3. Because the front and the rear are guard frequency, they are not used for transmitting signal. For samplings from 150 to 750, we can find that only in a few frequency the error is very large, and in other positions the error are very small.

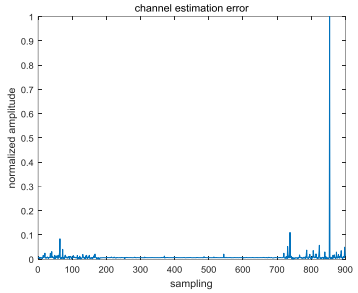


Fig. 3 channel estimation results

B. Communication Results

After estimation, equalization is carried on to the echo signal, and then decode to calculate bit-error-rate. We carry on 5 simulations to compare the ber of different SNR. The ber result is shown in Fig. 4. We can find that the bit-error-rate is decreasing as the SNR increases. When the SNR is 20dB, the ber is only about 0.3%. This is suitable for 2DPSK signal.

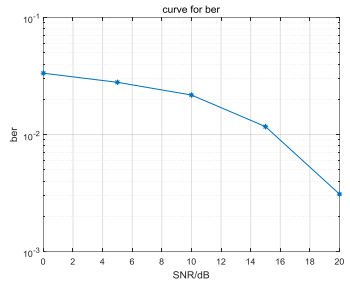


Fig. 4 bit error rate

C. Radar Results

After decoding communications, we can reconstruct reference signal by symbols we recovered. At procedure of pulse compression, the result is shown in Fig. 5.

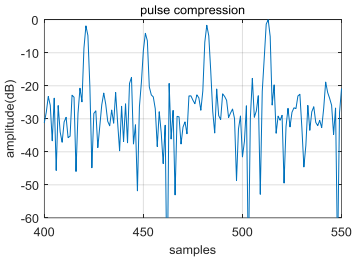


Fig. 5 pulse compression result

In Fig. 5, peaks are located at 420,453,481,514. According to sampling frequency and pulse duration time, we can find that the position is correspond to the scenario that we set in the front of this section. And the resolution of signal can also satisfy the condition. Then we contrast phase of adjacent pulse in Fig. 6.

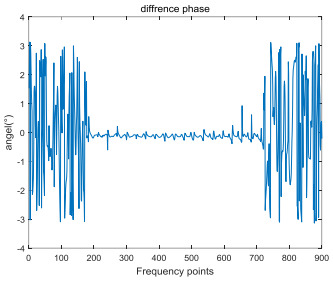


Fig. 6 difference phase

In Fig. 6, we can find that from 200 to 700, the phase difference is less than $\pi/4$ at the majority position of frequency. So we can take a further procedure to generate image of scenario. After imaging, the result is shown in Fig. 7.

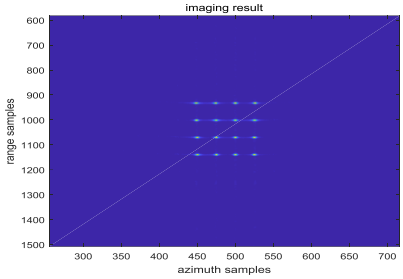


Fig. 7 imaging result

V. CONCLUSION

In this paper we use OFDM technique to combine the communication and radar system together, meanwhile we can get good results both in radar and communication. We first analyze the model of integration system and dig the constraints of signal generation. Via generated signals, we deduce the signal procedure models for both systems. And the results shows possibility of integration via this method proposed in this paper.

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